

Chapter 101 — Surgical, Radiographic, and Endoscopic Anatomy of the Female Pelvis

- The female pelvis houses **three systems** — **urological, reproductive and gastrointestinal** — and the urological surgeon must know all three, because the relationships between them determine where injury occurs.
- Anatomy is organised by **surgical approach** — **transvaginal, transabdominal and endoscopic** — followed by imaging (ultrasound and magnetic resonance imaging, MRI).
- Rationale: **anatomical knowledge prevents complications** and underpins good outcomes in both benign and malignant pelvic surgery.

TRANSVAGINAL APPROACH TO THE FEMALE PELVIS

- Order of description: **external soft tissue** → **bony pelvis (foundation)** → **musculature, nerves and vessels**.
- Structures must be understood in **lithotomy (the operating position)** and in their functional **standing or sitting position**.

External Genitalia

- **Vulva** = mons pubis, labia majora, labia minora, clitoris and vestibule.
- **Labia majora**: most superficial skin folds; meet anteriorly at the **mons pubis** (fat pad over the symphysis pubis, covering most of the clitoris).
- **Labia minora**: lie within the labia majora, more prominent anteriorly; fuse around the glans as **prepuce (anterior, hood-like fold)** and **frenulum (posterior fold)**; join posteriorly at the **fourchette**, usually torn at vaginal delivery.
- **Vestibule**: the cleft between the labia minora, into which the vagina and urethral meatus open.
- **Introitus**: anteroposterior slit fringed by the hymen. **External urethral meatus lies 1 cm anterior to the vaginal opening**.
- **Skene's (paraurethral) glands**: right and left ventral aspect of the meatus; ducts drain to the vulva; lubricate the meatus, secretion possibly antibacterial. **Embryological homologue of the prostate**; mature secretory and basal cells make it a functional genitourinary organ.
- **Perineal membrane is pierced by the vagina and urethra**.
- **Erectile tissue** (crura and vestibular bulbs) attaches to the inferior surface of the perineal membrane, covered as a pair: **crura** → **ischiocavernosus**; **vestibular bulbs** → **bulbospongiosus**.
- **Crura** attach to the medial ischiopubic rami → form the clitoral shaft → turn downward and backward → capped by the glans.
- **Vestibular bulbs** surround urethral and vaginal openings, fuse in the midline at the glans; the posterior end of each bulb expands over the **greater vestibular (Bartholin's) gland** — site of **Bartholin's cyst or abscess**, requiring surgery.

Perineum

- **Two triangles**, divided by a line between the ischial tuberosities: **anterior urogenital triangle** (pubic symphysis, pubic and ischial rami, deep transverse perineal muscles) vs **posterior anal triangle** (contains the anus).

- **Perineal membrane** (formerly **urogenital diaphragm**): fascial sheet across the pubic arch; urethra and vagina pass through its **urogenital hiatus** to the vestibule. It divides the urogenital triangle into deep and superficial spaces.

DEEP VS SUPERFICIAL PERINEAL SPACE

	Deep space	Superficial space
Boundary	Floored by the inferior fascia of the urogenital diaphragm (perineal membrane)	Roofed by Colles fascia
Contents	Urethral sphincter, urethrovaginal sphincter, compressor urethrae, deep transverse perineal muscles	Superficial perineal muscles, clitoris, vestibular bulbs, Bartholin's glands

- The table is testing **which side of the perineal membrane a structure lies**: sphincteric striated muscle deep; erectile tissue and Bartholin's glands superficial.
- **Vestibular bulbs** (corpus spongiosum tissue): two elongated erectile bodies either side of the urethra, close to the labia minora.
- **Colles fascia**: laterally fixed to the fascia lata at the ischiopubic ramus (**defines the perineal thigh crease**); apex at the arcuate ligament of the pubis; posteriorly wraps the superficial transverse perineal muscle and fuses with the perineal body; anteriorly covers the clitoris, **analogous to dartos in the male**.
- **Deep perineal fascia** overlies the superficial perineal muscles and fuses with the suspensory ligaments of the clitoris and the rectus sheath and external oblique fascia.
- **Perineal body**: central point of the perineum (muscle, collagen, elastic fibres); **convergence of bulbospongiosus, external anal sphincter, and superficial and deep transverse perineal muscles**; attached to the posterior border of the perineal membrane, between vestibule and anal canal.
- Obstetric relevance: perineal body damage at delivery injures the **external anal sphincter** → **episiotomy is angled laterally** to spare those fibres.
- **Bulbospongiosus**: splits around the introitus, attaches to the clitoris anteriorly and perineal body posteriorly, covers the bulbs; contraction **constricts the vaginal orifice and expresses vestibular gland secretions**.
- **Ischiocavernosus** covers a clitoral crus and **promotes clitoral erection**. **Superficial transverse perineal muscles** overlie the posterior vagina.

KEY POINTS: EXTERNAL GENITALIA AND PERINEUM

- The **vulva** describes the mons pubis, labia majora and minora, clitoris and vestibule.
- The crura of the clitoris and the vestibular bulbs are covered by the **ischiocavernosus and bulbospongiosus** respectively.
- The **perineal body** is the convergence of bulbospongiosus, external anal sphincter, and superficial and deep transverse perineal muscles.

Bony Pelvis

- The bony pelvis **supports the abdominal and pelvic viscera**.
- An oblique line from the **sacral promontory to the superior margin of the pubic symphysis** (the pelvic brim, defining the **pelvic inlet**) separates **true pelvis (below the brim)** from **false pelvis (above the brim)**.
- **True pelvis = four bones**: two hip (innominate) bones, sacrum and coccyx. Each innominate = **ilium + ischium + pubis**.
- **Ilium**: fan-shaped, wing and body. **Ischium**: fused to the ilium.

- **Ischial spines** project posteromedially at about the level of S5.
 - Located in lithotomy by palpating the sacrum → following the sacrospinous ligament laterally → ischial spine.
 - The **interspinous line divides the true pelvis into anterior and posterior compartments.**
 - **Key landmark in prolapse surgery** for a safe site of apical suspension sutures.
- **Pubis:** body + superior and inferior rami. Inferior ramus joins the ischium; superior ramus joins the ilium. **Ischiopubic ramus = lower margin of the obturator foramen;** superior pubic ramus and ischium form its upper margin. Pubic bones meet at the **pubic symphysis.**
- **Sacrum:** five fused vertebrae. The **S1 foramina fan out laterally relative to S4** — useful when **identifying S3 for sacral neuromodulation lead placement.**
- **Sacral promontory** is overlain by a **neurovascular bed** — **avoid during sacrocolpopexy mesh fixation;** bleeding here is hard to control.
- **Coccyx:** attached inferiorly to the sacrum; posterior border of the pelvic outlet.
- **Standing position:** anterior superior iliac spine and pubic symphysis lie in the same vertical plane; the **pelvic inlet faces anteriorly**, so intra-abdominal and pelvic pressure is **directed onto bone rather than onto muscle and fascia.** Surgery is described in lithotomy, but upright form and function determine success.

FEMALE VS MALE BONY PELVIS

Feature	Female	Male
Overall size	Larger, broader, sides wider apart	Narrower
Inlet	Oval and wider	Heart-shaped
Subpubic angle (between inferior rami)	Obtuse	Acute
Sacrum	Shorter, wider; less pronounced promontory	Longer, narrower; prominent promontory

- The pattern: every female difference **widens the outlet for parturition;** the wide oval inlet is the one the chapter links to **weakness of the pelvic floor.**

Ligaments of the Bony Pelvis

- Ligaments provide **stability of the bony pelvis.**
- **Sacrospinous ligament (SSL):** ischial spine (laterally) → side of the lower sacrum and coccyx (medially). **Commonly used to anchor apical suspension sutures** in prolapse repair.
- **SSL + sacrotuberous ligament** convert the sciatic notches into the greater and lesser sciatic foramina:

GREATER VS LESSER SCIATIC FORAMEN

Greater sciatic foramen	Lesser sciatic foramen
Piriformis; vessels and nerves to the gluteal muscles and posterior leg; sacral plexus exits here (immediately posterior to SSL); pudendal nerve leaves the pelvis here	Pudendal nerve and vessels (re-entering towards Alcock's canal); obturator internus

- **Cooper's ligament:** medial surface of the superior pubic ramus; **point of vesicourethral suspension in the Burch urethropexy** for stress urinary incontinence (SUI).
- **Obturator foramen:** covered by the obturator membrane except at the **obturator groove** (inferior surface of superior pubic ramus) → **obturator canal**, transmitting obturator vessels and nerve to the adductor compartment.

- **Obturator canal lies superolateral in the foramen** — critical when passing **transobturator sling trocars**: injury causes **debilitating groin pain and/or significant bleeding**.
- Obturator internus and externus overlie the foramen and membrane.

KEY POINTS: BONY PELVIS

- The **pubic rami, ischial spines and sacrum** are the main anchoring points for the ligaments supporting the pelvis.
- Standing, forces are **dispersed to the bony pelvis**, minimising stress on pelvic organs, muscles and fascia.
- The **SSL** (ischial spine → lower sacrum and coccyx) is often used as an attachment site in vaginal prolapse surgery.

Concept — The bony pelvis is read as a set of **fixation points and danger zones**: ischial spine, SSL, Cooper's ligament and sacral promontory are where sutures go, and each has an adjacent structure (pudendal nerve, sacral plexus, obturator canal, presacral vessels) that defines how close the surgeon may go.

Pelvic Floor Musculature and Fasciae

- **Pelvic walls**: obturator internus, iliacus, psoas major and minor, levator ani complex and coccygeus.
 - **Obturator internus**: most of the lateral side wall → through the **lesser** sciatic foramen → greater trochanter.
 - **Piriformis**: pads the posterolateral wall → through the **greater** sciatic foramen → greater trochanter. Sacral plexus lies medial to it.
- **Pelvic floor = pelvic diaphragm**, pubis to coccyx; **levator ani is its largest component**, crucial for support and function of the urogenital viscera.
- **Levator ani = puborectalis + pubococcygeus + iliococcygeus**, each named by its attachments; direct innervation from the **S3–S5 roots** (principally **S4**).

COMPONENTS OF THE PELVIC DIAPHRAGM

Muscle	Origin → insertion	Function / relevance
Puborectalis	Back of symphysis and superior fascia of urogenital diaphragm → loops behind the anorectal junction, joining its partner	U-shaped sling pulling the anorectal junction towards the pubis
Pubococcygeus	Posterior pubis and arcus tendineus → visceral organs and anococcygeal raphe	Visceral control; slings the vagina and urethra as the pubovaginal muscles
Iliococcygeus	Arcus tendineus levator ani and ischial spine → anococcygeal raphe and coccyx	Posterior, flat part of the levator plate
Coccygeus (ischiococcygeus; not part of levator ani)	Ischial spine → last two sacral segments and coccyx	Lies anterior to (and blends with) the SSL

- What the table tests: **which muscle forms which sling** — puborectalis the anorectal sling, pubococcygeus the urethrovaginal sling — and that **coccygeus is a separate muscle lying on the SSL**.
- **Levator hiatus**: opening transmitting urethra, vagina and rectum. **Levator plate**: midline fusion of levator ani, a **horizontal shelf on which the viscera rest**.
- **Causal chain of prolapse**: levator weakening → **wider levator hiatus** → **pelvic organ prolapse**.
- **Transvaginal assessment**: levator ani palpable through the lateral vaginal walls — tenderness or hypertonicity (pelvic floor dysfunction), and ability to contract and relax.

- **Transvaginal access to the retropubic space:** incise the anterior vaginal wall either side of the urethra (retropubic sling or transvaginal prolapse repair); the **arcus tendineus** is felt as a thick band of pelvic fascia from **ischial spine to pubis** — a fascial attachment point.
- **Resting tone** of the pelvic floor supports the viscera, resists rises in intra-abdominal pressure and provides **passive urinary and faecal continence**. Loss of tone (muscle or nerve injury) → lax urogenital hiatus → **levator plate loses its horizontal orientation**.
- The **International Continence Society (ICS)** has published updated terminology for pelvic floor muscle assessment to standardise communication and research.

KEY POINT: PELVIC FLOOR MUSCULATURE AND FASCIAE

- Sustained **resting tone** supports the viscera, resists raised intra-abdominal pressure and is crucial to **passive urinary and faecal continence**.

Concept — The pelvic floor works as a **horizontal levator plate under tonic contraction**; prolapse follows from anything that **widens the hiatus or tilts the plate**, which is why muscle and nerve injury converge on the same clinical result.

Vasculature and Lymphatic Drainage of the External Genitalia

- **Dual arterial supply to the labia majora:** superiorly — **superficial external pudendal artery** (from the femoral artery) vs inferiorly — **posterior labial branches of the internal pudendal artery** (from the internal iliac).
- This dual supply makes the labial fibrofatty tissue — the **Martius flap** — an ideal **interposition flap in vaginal reconstruction**.
- **Venous drainage:** external pudendal vein → great saphenous → femoral vein; internal pudendal vein → internal iliac vein.
- **Lymphatics:** vulva → **superficial inguinal** → **deep inguinal** → **pelvic nodes**. Clitoris and labia minora → deep inguinal, possibly to internal iliac nodes.

Concept — The external genitalia sit at the **junction of limb and pelvic territories** — femoral and internal iliac arteries, saphenous and internal iliac veins, inguinal then pelvic nodes — and the Martius flap exploits exactly that double supply.

Innervation to the External Genitalia and Pelvic Floor Musculature

- **Somatic** (sensory and motor: external genitalia and pelvic floor) vs **autonomic** (sympathetic and parasympathetic: pelvic organs).
- **Parasympathetic stimulation** → increased vaginal secretion, clitoral erection, engorgement of erectile tissue.

Somatic Innervation

- **Sacral plexus:** ventral rami **L4–L5 and S1–S4**; lies on piriformis, deep to endopelvic fascia, posterior to the internal iliac vessels; supplies posterior thigh and leg.
- Exits via the **greater sciatic foramen immediately posterior to the SSL** → **at risk during SSL fixation**.
- **Pelvic floor muscles:** direct innervation from **S3–S5**; the **pelvic somatic nerves (L4–S4)** travel on the pelvic surface of levator ani, supplying it and the **striated urethral sphincter**.
- **Pudendal nerve (S2–S4):** arises above the sacrotuberous ligament and coccygeus → leaves via the greater sciatic foramen → crosses piriformis, coccygeus and the **SSL close to the ischial spine (vulnerable in SSL**

fixation) → medial to the internal pudendal vessels through the lesser sciatic foramen → **pudendal (Alcock's) canal**.

- **Pudendal branches:** inferior rectal nerve; perineal nerve; dorsal nerve of the clitoris.
 - **Perineal nerve** → posterior labial branch (labium majus), superficial and deep transverse perineal muscles, **external urethral and anal sphincters**, levator ani. Pudendal fibres carry motor efferents to the pelvic floor and proprioceptive and sensory afferents from the urethra.
- **Pudendal nerve block** (unilateral or bilateral; treats **pudendal neuralgia**): anaesthetic into the **SSL near its ischial insertion, posterior to the ischial spine**, catching the nerve as it enters the lesser sciatic foramen; accessible **transvaginally**.
- **Posterior femoral cutaneous nerve (S1–S3):** through the greater sciatic foramen; perineal sensory branch (**sensory innervation of the perineum**); lateral perineal skin also via its perineal branch.

NERVES OF THE FEMALE PELVIS AND PERINEUM AT SURGICAL RISK

Nerve (roots)	Territory	Mechanism or procedure of injury	Deficit
Common peroneal (from sciatic)	Tibialis anterior; lower leg	Lithotomy — compression at the fibular head , where it is superficial	Foot drop (loss of dorsiflexion)
Sacral plexus (L4–S4)	Posterior thigh and leg	SSL fixation (exits just posterior to SSL)	Motor and sensory loss, posterior leg
Pudendal (S2–S4)	Perineum, sphincters, clitoris	SSL fixation near the ischial spine	Perineal pain and sensory loss; sphincter dysfunction
Sacral nerve trunks (S1, S2–S4)	—	Uterosacral suspension (entrapment dorsally)	Buttock and posterior thigh pain
Ilioinguinal (L1)	Mons, anterior labium majus	Inguinal hernia repair, open appendicectomy, caesarean section, total abdominal hysterectomy	Mons and labial sensory loss or pain
Genitofemoral (L1–L2)	Mons and labia	Inguinal herniorrhaphy; standard and extended lymphadenectomy	Genital sensory loss or pain
Posterior femoral cutaneous (S1–S3)	Perineum, posterior thigh	Gluteal pressure injury; pelvic tumours	Perineal sensory loss

- What this table is for: **matching the injured nerve to the operation**. The trap is attributing post-lithotomy foot drop to a pelvic cause — it is **peripheral compression of the common peroneal nerve at the fibular head**.
- **Perineal nerve branches supply the posterior two-thirds of the labia**; ilioinguinal supplies the anterior labium.
- **Compartment syndrome after lithotomy:** prolonged pressure on posterior leg muscles → necrosis → oedema → raised compartment pressure → vascular compromise. **A surgical emergency**; suspect in **postoperative leg pain and swelling**. **Intermittent external compression devices** may reduce pressure rises in lithotomy.

KEY POINTS: INNERVATION OF THE EXTERNAL GENITALIA AND PELVIC FLOOR

- Cutaneous innervation of the mons and labia: **ilioinguinal (L1) and genitofemoral (L1–L2)**.
- Sensory innervation of the perineum: **posterior femoral cutaneous nerve (S1–S3)**.
- The **sacral plexus exits immediately posterior to the SSL** and can be injured in SSL fixation.

- For pudendal neuralgia, the pudendal nerve can be **blocked transvaginally**.

Concept — *Innervation is taught by the operation that endangers it: SSL fixation threatens the pudendal nerve and sacral plexus, groin and lower abdominal surgery the L1 nerves, and lithotomy the common peroneal nerve — the ischial spine being the single landmark around which the pelvic nerve risks cluster.*

Relevant Thigh Anatomy

- **Fascia lata** grafts: **autologous pubovaginal slings**, and **in place of mesh in sacrocolpopexy**.
- **Function**: stabilises the knee, confines thigh muscle expansion (more efficient contraction), aids venous return. Inferiorly attaches to distal femur and tibia; laterally forms the **iliotibial band**.
- **Safe harvest**: identify the **lateral epicondyle** (just lateral to the patella) → **avoid harvest near the condyle**, where the band is narrow → preserve the band's longitudinal integrity.

TRANSABDOMINAL APPROACH TO THE FEMALE PELVIS

Pelvic Viscera

Urologic Organs

Urethra.

- **Length 4.7–5.1 cm**; runs behind the pubic symphysis to the bladder neck. Suspended by the **posterior pubourethral ligaments** and anteriorly by the **suspensory ligaments of the clitoris**.
- **Muscle wall**: **inner longitudinal and outer circular smooth muscle**, beneath a mucosa continuous with the bladder; longitudinal fibres **shorten and widen the urethra during voiding**. Elastic tissue and collagen surround the layers.

INTERNAL VS EXTERNAL URETHRAL SPHINCTER

	Internal sphincter	External sphincter
Muscle	Smooth — continuation of the detrusor	Striated
Control	Involuntary (autonomic)	Voluntary
Site	Proximal urethra and bladder neck	Distal two-thirds ; horseshoe-shaped at its proximal (mid-urethral) part
Relation	—	Anatomically separate from periurethral striated muscle of the anterior pelvic floor

- Components of the **striated urogenital sphincter complex**:
 - **Urethral compressor**: lateral fibres continuous with the anterolateral vaginal walls → **closes the urethra against the anterior vaginal wall**.
 - **Urethrovaginal sphincter**: fibres encircling urethra and vagina → **narrows the urogenital hiatus**.
 - **Pubococcygeus** runs bilaterally along the urethra → increases urethral resistance.
- **Periurethral glands**: obstruction → **urethral diverticulum or periurethral cyst**. Most prominent: **Skene's glands** (prostate homologue), opening just inside the meatus.
- **Layers**: epithelium → submucosa (lamina propria with a spongy vascular network) → muscle; surrounded by a thin **periurethral fascia**.
- **Epithelium**: transitional, with infoldings for distensibility and **coaptation**; becomes pseudostratified then squamous distally.
- Pathogenesis of diverticulum: **infected or ruptured paraurethral gland disrupting the periurethral fascia**.
- **Mucosa and submucosa generate most urethral closure pressure and are oestrogen-dependent**.

- **Orientation** as in the male: **ventral = against the vagina; dorsal = against the pubis.**
- **Bladder neck and urethral hypermobility** is one cause of female SUI.

SLING POSITION

Autologous fascial pubovaginal sling	Synthetic mid-urethral sling
Bladder neck	Mid-urethra

Bladder.

- **Extraperitoneal** hollow reservoir behind the pubic symphysis; anterior surface abuts the **retropubic space of Retzius.**

BLADDER SURFACES AND THE CONSEQUENCE OF INJURY

Surface	Covering / relation	Consequence of injury
Superior posterior (dome)	Peritoneum	Intraperitoneal leak → surgical repair: wall thinner, heals poorly with drainage alone; bowel may herniate through the defect
Inferior and inferolateral	Endopelvic fascia	—
Inferoposterior (base)	Anterior vaginal wall and lower uterus	Vesicovaginal or vesicouterine fistula
Bladder neck	Pubourethral ligaments; pelvic floor below	—

- Otherwise **mobile**, to accommodate filling. In females, the **proximal urethra and bladder neck function as the internal sphincter**; urethra and bladder neck also rest on the anterior vaginal wall.
- Wall: **urothelium → lamina propria → muscularis propria (detrusor).**

Ureter.

- Retroperitoneal. **Abdominal ureter:** kidney to iliac vessels. **Pelvic ureter:** about **half the total length**, from the iliac crossing to the bladder.
- **Course (sequence):** crosses the pelvic brim over the **common iliac vessels near their bifurcation** → along the lateral pelvic wall towards the ischial spine → **posterior to the ovary** → through the **base of the broad ligament** → **under the uterine vessels ("water under the bridge")** → turns medially at the ischial spine → **over the anterior vaginal fornix** → bladder.
- Injury sites: **under the uterine artery at hysterectomy**; at the **anterior vaginal fornix in transvaginal surgery.**

Female Reproductive Organs

- **Uterus:** normally **anteverted and anteflexed**; body + cervix.
- **Cervix:** about **2.5 cm**, adjacent to the bladder, separated by parametrium; opens as the os. **Fornices:** **shallow anterior (ventral), deeper posterior (dorsal),** and lateral.
- **Uterine wall (outside → in):** perimetrium (peritoneum and thin connective tissue) → myometrium (outer longitudinal, continuous with ovarian and round ligaments; **middle circular, carrying the vessels and nerves;** inner longitudinal) → endometrium.
- **Fallopian tubes:** **10–12 cm**, draped in the broad ligament; **interstitial → isthmus → ampulla → infundibulum** → fimbriae; enter the uterus at the cornua.
- **Ovary:** in the **ovarian fossa** (bounded by obliterated umbilical artery, ureter and internal iliac artery), held by the mesovarium to the posterior broad ligament; suspended by the **infundibulopelvic ligament (carries the ovarian vessels);** tethered to the uterus by the **ovarian ligament.**

- **Vagina:**

- **Columnae rugarum** along anterior and posterior walls, ending at the urethrovaginal ridge (carina); **H-shaped on cross-section**, with rugae.
- Mucosa (**non-keratinised stratified squamous**) and lamina propria fixed to muscle (outer longitudinal, inner circular), attached bilaterally to the surrounding endopelvic fascia.
- The vagina attaches **higher on the cervix posteriorly**, so **anterior wall about 7.5 cm vs posterior wall about 9 cm**.
- Attached anteriorly via levator ani to the **arcus tendineus**, posteriorly to the **rectovaginal septum**.
- Bladder base is tethered to the vagina by smooth muscle fibres, **divided to enter the vesicovaginal space** (vesicovaginal fistula repair).
- Ureters pass close to the **lateral fornices** and are **anterior to the vagina** as they enter the bladder.

Gastrointestinal Organs

- **Descending colon (retroperitoneal) → sigmoid** at the iliac crest; the sigmoid gains a longer mesentery and becomes **intrapertitoneal**; fixed iliac and mobile pelvic portions.
- **Left ureter runs through the base of the sigmoid mesocolon**; landmark: **intersigmoid fossa**.
- **Rectosigmoid junction is disputed**: at the **sacral promontory** vs more distally at the **narrowest point where the taeniae fuse**.
- **Rectum**: retroperitoneal, **12–15 cm, no taeniae or haustra**; becomes the anal canal after passing through levator ani.

KEY POINTS: PELVIC VISCERA

- The **ureter's relationship to the reproductive organs** makes it a common site of injury in gynaecological surgery.
- The **rectum** is retroperitoneal and differs from colon in having **no taeniae or haustra**.

Concept — Every pelvic organ is described by **what it rests on**: bladder base, urethra and bladder neck all lie on the anterior vaginal wall, and the ureter threads between uterine vessels and vaginal fornix — so **continence, fistula and ureteric injury are all problems of the anterior vaginal wall and its neighbours**.

Support Structures for the Pelvic Viscera

Fascia and Peritoneum

THE THREE FASCIAL STRATA

Stratum	Components	Significance
Inner	Rectal fascia over the anterior and lateral rectal wall, vessels and nerves	Forms part of Denonvilliers fascia
Intermediate	Retroperitoneal connective tissue around uterus and vessels: pubovesical and pubocervical fascia around the vagina; parametrium (on the uterus); paracolpium (around the vagina)	Provides most of the support of the pelvic organs
Outer	Transversalis fascia (continuous with endopelvic and lateral pelvic fascia); iliac fascia (over iliacus and psoas, iliac crest → tendinous arch, the "white line")	Endopelvic fascia runs from the uterine artery to the vagina–levator fusion; key at organ exit points

- The table's point: **support comes from the intermediate stratum**, and **parametrium vs paracolpium** is simply uterus vs vagina.

- Iliac fascia is continuous with the inguinal ligament, pectineal and obturator fascia; **obturator fascia** covers obturator internus and piriformis.
- **Inferior pelvic fascia:** continuous with obturator fascia and pudendal canal fascia; covers levator ani.
- **Superior pelvic fascia:** from the outer stratum and obturator fascia; pubic symphysis → ischial spine. Fascia is thinner over muscle and organs, **permitting mobility**.

Pelvic Ligaments (Visible Transabdominally)

- Pubovesical, vesicopelvic, cardinal and uterosacral ligaments and pubocervical fascia are **fascial condensations** supporting the pelvic organs.

PELVIC LIGAMENTS: COURSE, CONTENTS AND SURGICAL RELEVANCE

Ligament	Course / attachments	Contents	Relevance
Pubovesical (pubourethral)	Pubis → bladder neck	—	Homologue of puboprostatic ligaments; retropubic suspension; hammock under the mid-urethra with the anterior vaginal wall; weakness → SUI
Vesicopelvic (fused perivesical and endopelvic fascia)	Bladder base and anterior vaginal wall → tendinous arch; continuous with periurethral fascia, cervix and cardinal ligament	—	Defect → lateral (paravaginal) cystocele
Broad	Peritoneal fold from uterus to pelvic side wall	Fallopian tube, ovary; uterine vessels and nerves (mesometrium); mesosalpinx	—
Round	Lateral uterine body → internal inguinal ring → over external iliac artery → labium majus	—	Homologue of the gubernaculum
Infundibulopelvic (suspensory ligament of ovary)	Ovary → pelvic side wall , behind the broad ligament	Ovarian vessels	Ureter lies close beneath it
Ovarian	Ovary → uterus	—	—
Cardinal (transverse cervical, Mackenrodt's)	Pelvic side wall → posterolateral pericervical fascia and lateral vaginal wall; fuses posteriorly with uterosacral	Major vessels from the internal iliac; pelvic plexus nerves	Stabilises uterus, cervix, upper vagina; hysterectomy injury → bladder dysfunction
Uterosacral (sacrouterine)	Sacrum (S2–S4) → lateral fascia encircling cervix, uterine isthmus and vaginal wall	Fibrous tissue, smooth muscle; pelvic plexus nerves	Anchor for apical suspension (vaginal or abdominal); ureter lateral to its anterior part, closest at the cervix ; risk of sacral nerve entrapment
Anterior longitudinal (ALL)	Anterior vertebral column; 1.3–2.5 mm thick	—	Sacrocolpopexy anchor

- What the table tests: **which ligament carries which vessels** (infundibulopelvic → ovarian; cardinal → internal iliac branches, uterine artery) and **which ligaments are suspension anchors** (uterosacral, sacrospinous, ALL).

- The classic confusion: **infundibulopelvic (ovary to side wall, with ovarian vessels) vs ovarian ligament (ovary to uterus).**
- The **ureter** is endangered where it passes the ovarian and uterine supports: beneath the infundibulopelvic ligament and ovarian vessels, **under the uterine artery**, and alongside the cardinal ligament close to the cervix.

Sacrocolpopexy dissection.

- Sequence: expose the ALL at the **sacral promontory over the L5/S1 disc** → dissect peritoneum distally → secure sutures to the ALL over the **anterior surface of S1.**
- Why S1: **lower risk of discitis or osteomyelitis**, and the **highest tensile strength** of the ALL.
- Hazards: **left common iliac vein, middle sacral artery and vein, presacral venous plexus, right ureter.** In cadavers, the right ureter and left common iliac vein each lie about **2.7 cm** from the mid-promontory.

KEY POINTS: SUPPORT STRUCTURES FOR THE PELVIC VISCERA

- The **infundibulopelvic ligament** runs from the ovary to the pelvic side wall behind the broad ligament and **contains the ovarian vessels.**
- The **uterosacral ligaments** run from the sacrum to the fascia around the cervix, uterine isthmus and vaginal wall and are used in **vaginal apex suspension.**
- The **ALL** overlies the anterior vertebral column and is the **suture site in sacrocolpopexy.**

Concept — Pelvic support is **connective tissue condensed around vessels and nerves**; that is why the same ligaments that make good suspension anchors (uterosacral, cardinal) are the ones whose division or suturing injures the ureter, pelvic plexus and sacral nerves.

Vasculature and Lymphatic Drainage of Pelvis

Arterial Supply

- **Aorta** enters the retroperitoneum through the diaphragm at **T12** → bifurcates at the **left anterolateral L4** into common iliac arteries.
- **Common iliac arteries: right about 5 cm, left about 4 cm** → bifurcate into internal (hypogastric) and external iliac arteries **where the ureter crosses anteriorly.**
- **Middle sacral artery:** arises at the aortic bifurcation, runs **posterior to the left common iliac vein**, joined by the middle sacral vein.
- **External iliac:** along iliopsoas; main supply to the lower limb (via femoral) and abdominal wall (via inferior epigastric). Its femoral branch gives the **external pudendal artery** (labia majora).
- **Primary supply of the external genitalia** (labia minora, clitoris, vestibule) is the **internal pudendal artery**, from the internal iliac.
- **Internal iliac:** supplies pelvis and viscera; divides at the **upper border of the greater sciatic foramen.**

INTERNAL ILIAC ARTERY: ANTERIOR VS POSTERIOR DIVISION

	Anterior division	Posterior division
Branch type	Parietal and visceral	Parietal only
Visceral	Obliterated umbilical, superior vesical, uterine , vaginal, middle rectal	—

	Anterior division	Posterior division
Parietal	Obturator, internal pudendal, inferior gluteal	Iliolumbar, lateral sacral, superior gluteal

- The table's trap: the **posterior division gives no visceral branches**; every organ artery arises anteriorly.
- **Branching is highly variable**: the obturator artery may arise from the internal or external iliac system (including from the inferior epigastric), and on meta-analysis the **uterine artery origin is inconsistent** — surgeons need general knowledge plus adaptability.
- **Collaterals**: uterus ← contralateral uterine (arcuate arteries) and **ovarian arteries** (from the aorta, below the renal arteries); bladder ← contralateral superior vesical and vaginal arteries.

COLLATERAL RESERVE: INTERNAL VS EXTERNAL ILIAC TERRITORY

Internal iliac	External iliac
Robust bilateral collaterals → embolisation or ligation for haemorrhage is usually tolerated	Poor collaterals → inadvertent ligation threatens the limb; a surgical emergency

Venous Drainage

- Pelvic viscera → **internal and external iliac veins** → **common iliac veins** → **inferior vena cava (IVC)**.
- **Exception** — **ovarian veins** (in the infundibulopelvic ligaments): **right** → IVC; **left** → **left renal vein**.
- External and internal iliac veins join **anterior to the sacroiliac joint**; the common iliac veins unite at about **L5**.
- **Median sacral vein** → left common iliac vein or the confluence; met during promontory dissection.
- **External iliac vein** = continuation of the femoral vein (behind the inguinal ligament); receives deep circumflex iliac, inferior epigastric and **pubic veins** (the pubic vein links external iliac and obturator veins).
- **Internal iliac vein: posteromedial to the artery**; tributaries even less predictable than the arteries, with **duplicated internal iliac veins common**.
 - Parietal: gluteal, internal pudendal, obturator, **lateral sacral veins**.
 - Visceral: middle rectal, vesical, uterine, vaginal.
- **Sacral venous plexus: valveless** network on the anterior sacrum (median and lateral sacral veins, posterior abdominal wall and basivertebral veins). Injured in presacral dissection — **sacrocolpopexy, radical hysterectomy, abdominoperineal and low anterior resection**.
- High hydrostatic pressure, especially in lithotomy: **a 1–2 mm vein can bleed more than 1 L/min** → dissect with caution or **avoid the area when not needed**.

Lymphatic Drainage

- Normal nodes are **impalpable**, hidden in the intermediate stratum.
- **Groups**: common iliac, external iliac, internal iliac, inguinal, perivisceral — named for adjacent vessels or organs.
- **Most visceral drainage passes through the internal iliac nodes** and their branches: **presacral, obturator, internal pudendal**.
- Inguinal nodes communicate directly with internal and external iliac chains. **Common iliac nodes** receive external iliac, internal iliac and pelvic ureteric lymph → **lateral aortic nodes**.

EXTERNAL ILIAC NODAL CHAINS

Chain	Position	Drains
External	Lateral to vessels	Anterior abdominal wall, clitoris
Middle	Over the external iliac artery	Bladder, vagina

Chain	Position	Drains
Internal	Medial to vessels	Lower abdominal wall, inguinal nodes, bladder neck, urethra

LYMPHATIC DRAINAGE BY ORGAN

Organ	First drainage	Onward
Distal one-third urethra (anterior)	Superficial and deep inguinal	Pelvic nodes
Proximal two-thirds urethra (posterior)	Iliac and obturator	Common iliac
Upper two-thirds vagina	Deep pelvic nodes (as other viscera)	—
Lower one-third vagina, vulva	Inguinal	Pelvic nodes
Clitoris, labia minora	Deep inguinal	Internal iliac
Bladder	Perivesical fat	Pelvic → presacral → great vessel nodes
Uterus	Broad ligament	Pelvic → presacral → great vessel nodes
Cervix	Along uterine vein → internal iliac; along artery → internal and external iliac	—
Ovaries, tubes	Along gonadal vessels: left → renal vein region; right → IVC	Pelvic nodes may also receive

- The pattern: **the lower third of the tract (distal urethra, lower vagina, vulva) drains to the groin; the upper two-thirds drains to the pelvis.** The trap is the **ovary, which bypasses the pelvis** and follows its vessels to para-aortic territory.
- **Bladder and uterus drain analogously:** perivisceral → pelvic → presacral → great vessel nodes.

LEISSNER CLASSIFICATION OF EXTENDED PELVIC LYMPHADENECTOMY (BLADDER CANCER)

Level	Nodes
1	Internal and external iliac, obturator, perivesical
2	Common iliac, presacral
3	Paracaval, para-aortic, interaortocaval

- Pelvic lymphadenectomy removes nodes with **primary or downstream drainage of the organ**; the levels simply **follow the drainage cascade upward**.

KEY POINTS: VASCULATURE AND LYMPHATIC DRAINAGE OF THE PELVIS

- The **internal iliac artery** supplies the pelvis and viscera, dividing into anterior and posterior divisions at the upper border of the greater sciatic foramen.
- Pelvic visceral veins drain via internal and external iliac veins → common iliac → IVC.
- **Distal one-third urethra → inguinal nodes; proximal two-thirds → iliac and obturator nodes.**

Concept — Pelvic vascular anatomy is governed by **redundancy vs predictability**: the internal iliac territory is variable but richly collateralised (ligation tolerated), the external iliac is constant but end-arterial (ligation catastrophic), and the valveless presacral veins make the sacrum the one place where a small vessel produces exsanguinating bleeding.

ENDOSCOPIC ANATOMY

- **Cystourethroscopy sequence:** distal urethra (periurethral glands, mucosal folds) → **coaptation distally at the mid-urethra (external sphincter)** → **coaptation proximally at the bladder neck** → trigone and orifices → survey of the whole mucosa.
- **Urethral findings: foreign bodies (mesh, stones) and diverticular ostia.**
- **Ureteric orifices** along the **trigonal ridge**; if anatomy is unaltered (no previous surgery, disease, prolapse or radiation), seen with the scope at the bladder neck using a **30-degree lens**.
 - **Patulous orifice** → **vesicoureteric reflux**; **ectopic position** → **duplicated system**.
 - Significant anterior or apical descent: **sponge stick in the vagina** reduces the prolapse and restores the vesicourethral angle.
- **Landmarks on survey: symphysis pubis indentation** (anterior wall); **vessels and/or bowel** (lateral and anterolateral walls); **uterine impression** (proximal bladder floor and posterior bladder wall).
- **Normal findings:** rugae in the empty bladder that **flatten with filling**; delicate vasculature, denser on the trigone.
- **Benign trigonal (and sometimes lateral wall) lesions: squamous metaplasia, cystitis cystica** — not reliably diagnosed on appearance, so **suspicion of malignancy** → **biopsy**.
- **Post-radiation mucosa: friable and hypervascular.**

Concept — Female cystoscopy is interpreted against **the organs pressing on the bladder** — pubis anteriorly, bowel laterally, uterus posteriorly — and any deviation from the normal orifice shape or position, or any lesion not certainly benign, is a finding to act on rather than describe.

RADIOGRAPHY OF THE FEMALE PELVIS

Ultrasound

- **Translabial, transrectal and transvaginal ultrasound:** diagnosis of **pelvic organ prolapse** and **urethral disorders** (injury, stricture).
- **Translabial ultrasound** readily shows the pubic symphysis, urethra, bladder neck, vagina, rectum and anal canal.
 - **Levator plate and pubovisceral muscles:** hyperechoic area behind the anorectal junction. Cul-de-sac: fluid, fat or small bowel.
 - **Bladder neck mobility:** real-time imaging during **Valsalva, supine**. Also measures bladder wall thickness and prolapse.
 - Visualises **transvaginal mesh, mid-urethral slings, urethral bulking agents and implants**.
 - **Three-dimensional probes** add the **axial view** to the midsagittal plane.
- **Endoanal (anorectal) ultrasound:** anal sphincter defects in faecal incontinence.

ENDOANAL ULTRASOUND: INTERNAL VS EXTERNAL ANAL SPHINCTER DEFECT

Internal anal sphincter defect	External anal sphincter defect
Echogenic (bright) break in a hypoechoic muscle	Hypoechoic (dark) lesion in a normally echogenic muscle

- Spasm or hypertrophy (e.g. obstructed defaecation) → **thickened sphincter**.

Magnetic Resonance Imaging

- **MRI is the most comprehensive and conclusive radiation-free modality** for the female pelvis; contrast rests on differing water content (muscle, fat, fluid, blood).

- **Pelvic floor sequences:** T2-weighted **half-Fourier acquisition single-shot turbo spin-echo (HASTE)** or **single-shot fast spin-echo (SSFSE)** — rapid multiplanar surveys and **dynamic rest-and-strain studies**. Obturator internus and levator ani are well seen.
- **T2-weighted imaging** characterises masses, cysts and parenchyma (pelvis, urethra).
- **Benign vaginal cysts:** müllerian, epidermal inclusion, Gartner duct, Bartholin gland, Skene gland.
- **T2-weighted MRI is the best test to show and localise a urethral diverticulum** and distinguish it from other benign vaginal masses.

UTERINE ZONAL ANATOMY ON T2-WEIGHTED MRI

Zone	Signal
Endometrium and cavity	High
Junctional zone	Low
Outer myometrium	Intermediate to high

Magnetic resonance defecography.

- **Non-invasive dynamic study** of pelvic floor anatomy and function during straining and defaecation, with **rectal contrast**.
- Reference: the **pubococcygeal line (PCL)** — inferior border of the pubic symphysis → last coccygeal joint.
- Prolapse graded **mild, moderate or severe** per compartment (**anterior, middle, posterior**) by **maximum descent below the PCL**.
- Also shows **concomitant disorders**, e.g. urethral hypermobility.
- Most valuable in **defaecatory dysfunction** or when **symptoms are out of proportion to examination**.

KEY POINTS: RADIOGRAPHY OF THE FEMALE PELVIS

- **Translabial ultrasound** shows transvaginal mesh, mid-urethral slings, bulking agents and urethral implants.
- **T2-weighted MRI** is the best technique to visualise and localise urethral diverticula.
- **MR defecography** is valuable in prolapse with defaecatory dysfunction or symptoms out of proportion to examination.

Concept — Imaging is chosen by question: **ultrasound for implants and real-time mobility, static T2 MRI for soft-tissue characterisation** (diverticulum, cysts, uterine zones), and **dynamic MRI for function** measured against a fixed bony line.

FIGURES

- **Fig. 101.1** Female vulva and external genitalia (Gray's Anatomy, 40th ed., 2008).
- **Fig. 101.2** Female perineum; muscles removed on the left to show vestibular bulb and Bartholin's gland (Gray's Anatomy, 35th ed., 1972).
- **Fig. 101.3** Bony pelvis and ligaments, including sacral foramina, SSL and sacrotuberous ligament.
- **Fig. 101.4** Sagittal pelvis: false (major) vs true (minor) pelvis divided at the iliopectineal line (Textbook of Diagnostic Sonography, 2018).
- **Fig. 101.5** Bony pelvis standing: anterior-facing inlet.
- **Fig. 101.6** Muscles of the true pelvis.
- **Fig. 101.7** Arteries and nerves of the female perineum (Textbook of Gynecology, 1993).

- **Fig. 101.8** Urethra and supports: external urethral sphincter, urethral compressor, urethrovaginal sphincter (Hinman's Atlas, 2012).
- **Fig. 101.9** Fascia of parametrium and paracolpium, pelvic spaces and cardinal (Mackenrodt's) ligament with uterine artery.
- **Fig. 101.10** Broad ligament and contents; pubocervical fascia — horizontal part supports bladder and vagina, distal vertical part supports urethra and is the backstop against which it is compressed on straining.
- **Fig. 101.11** Ligaments and support of vagina and uterus; ovarian–uterine anastomosis.
- **Fig. 101.12** Relationship of the ureter to uterine structures.
- **Fig. 101.13** Sacral dissection: right common iliac artery, left common iliac vein, middle sacral vessels, right ureter at L5–S1.
- **Fig. 101.14** Arterial supply of the pelvis: anterior and posterior trunks (Gray's Anatomy, 40th ed., 2008).
- **Fig. 101.15** Venous drainage of the pelvis.
- **Fig. 101.16** Lymph vessels and nodes of the female pelvis (Sobotta Atlas, 2013).
- **Fig. 101.17–101.19** Cystoscopic views (30-degree lens): bladder neck, urethral sphincter, ureteric orifices, symphyseal, bowel and uterine impressions.
- **Fig. 101.20** Midsagittal translabial ultrasound: bladder, pubic symphysis, urethra, vagina.
- **Fig. 101.21** Endoanal ultrasound, normal: external and internal anal sphincters.
- **Fig. 101.22** Axial T2 MRI: bladder, vagina, rectum, obturator internus, levator ani, pectineus.
- **Fig. 101.23** Benign vaginal cystic lesions on MRI (Fundamentals of Body MRI, 2017).
- **Fig. 101.24** Sagittal T2 MRI: uterine zonal anatomy, fluid in the pouch of Douglas (Gray's Anatomy, 40th ed., 2008).
- **Fig. 101.25** MR defecography at rest and defaecation: generalised pelvic floor descent with moderate cystocele and rectocele, relative to the PCL.